

Frontline Shift Handover Copilot

Every step, start to finish, including the current (2026) way to build the tool in Step 6 — an agent flow writing to a new Dataverse table and notifying via Teams.

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What You're Building

An agent that captures shift handovers and safety incidents in conversation on hand-held devices, grounded in one data file, acting through one governed tool.

AGENT NAME	Frontline Shift Handover Copilot
ORCHESTRATION	Generative (AI-led)
KNOWLEDGE	Frontline_Shifts_Data.xlsx
CUSTOM TOPIC	Write Shift Handover Note
TOOL	Log Safety Incident (agent flow → Dataverse + Teams)

Files & Where They Go

FILE / ITEM	GOES TO	USED IN
Frontline_Shifts_Data.xlsx	Copilot Studio → Knowledge tab → Add knowledge → File	Step 3
SafetyIncidents_DataverseImport_Template.xlsx	Power Apps → Tables → New table → From Excel	Step 6
SafetyIncidentLogged_AdaptiveCard.json	Paste into the flow's Adaptive Card field	Step 6

Before you start — confirm Environment Maker permission, and Dataverse System Customizer (or higher) so you can create a new custom table.

Set Up the Agent

Steps 1–3 — create the agent shell, pick how it reasons, and ground it in data.

1 Create Your Copilot Studio Agent

Open Copilot Studio (copilotstudio.microsoft.com), select **Create** → **New agent**, then configure:

AGENT NAME	Frontline Shift Handover Copilot
DESCRIPTION	Helps shift leads write handover notes, log safety incidents, run pre-shift checks, and spot recurring near-misses across sites.
SOLUTION / ENVIRONMENT	Frontline Operations · Default environment
ICON	Use a recognisable colour and emoji or upload a brand icon

2 Choose Orchestration Mode

Under **Settings** → **Generative AI**, use **Generative Orchestration** (AI-led) so the agent can blend knowledge lookups with tool calls for shift handover and safety reporting.

3 Add Knowledge

Knowledge tab → **Add knowledge** → **File** → upload **Frontline_Shifts_Data.xlsx**. Contains:

- Shift metadata (shift ID, site, date, shift type — Early / Late / Night, lead)
- Operational data (head count, equipment check passed Y/N)
- Safety data (incident count, near-miss count, incident summary)
- Continuity (open actions, free-text handover notes)

You can layer additional sources later — SharePoint sites, Dataverse, public web, or Graph connectors.

Behavior: Instructions & Topic

Steps 4–5 — tell the agent how to behave, then give it a reusable conversation flow.

4 Add Agent Instructions

Overview → Instructions — paste exactly:

You are the Frontline Shift Handover Copilot. You help two audiences:

- 1) Shift leads writing their end-of-shift handover.
- 2) Site managers reviewing safety trends across sites.

Rules:

- Use the uploaded shifts dataset as your single source of truth.
- Keep answers short, plain-English, and action-led – these run on hand-held devices.
- For incident reports, ALWAYS call the 'Log Safety Incident' tool – do not say an incident is logged unless the tool returns success.
- Surface recurring near-misses (same site, same equipment, last 14 days).
- If an answer requires judgement on emergency response, escalate to a human.

5 Add a Custom Topic

Topics → Add a topic → Create from description:

Topic name: Write Shift Handover Note

Description: Captures a structured end-of-shift handover from the shift lead, stitches in equipment checks and any incidents, and stores the note against the shift row.

Trigger phrases:

- Write my handover note
- End of shift
- Log a handover
- Pass the shift to the next lead

Conversation steps:

- Identify shift (site, date, shift type) – confirm if multiple match
- Pull current row from knowledge – show head count, incidents, near-misses
- Ask: equipment check passed? any incidents? any open actions for next lead?
- Compose a 4-line plain-English handover note from the answers
- Read it back and ask the lead to confirm – edit if requested
- Call the 'Submit Shift Handover' tool to persist the note and notify the next lead

Build the Tool (Current Method)

Step 6 — updated for 2026, using current Copilot Studio tool-building.

Note on this step — the original deck's topic (Step 5) calls a tool named "Submit Shift Handover," but the deck's own Step 6 documents "Log Safety Incident" as the tool to build. This inconsistency exists in the source material itself. This guide builds what Step 6 actually specifies — Log Safety Incident.

6a Create the Dataverse Table First

In the **Power Apps maker portal** (make.powerapps.com) → Tables → **New table** → **Import from Excel** → upload **SafetyIncidents_DataverseImport_Template.xlsx**. This creates the table and columns from the workbook's headers in one step; rename the table **SafetyIncidents** and set **IncidentType** to a Choice column matching the Excel dropdown if it doesn't map automatically.

6b Build the Agent Flow

1. **Flows** → **New flow** → **Agent flow**.
2. Trigger inputs: **ShiftID**, **IncidentType**, **Location**, **Description**, **ActionTaken**.
3. Action: **Dataverse** — **Add a new row** → table **SafetyIncidents** → map the fields.
4. Action: **Teams** — **Post your own adaptive card as the Flow bot to a user** → the site Safety Officer → paste the contents of **SafetyIncidentLogged_AdaptiveCard.json** into the Adaptive Card field, then replace each {{Placeholder}} with the matching field from Dynamic content.
5. **Respond to the agent** → output **IncidentReference** from the new row's ID. Confirm **Asynchronous response = Off**.
6. **Publish** the flow.

6c Attach It as a Tool

1. **Tools** → **Add a tool** → filter by **Flow** → select it → **Add and configure**.
2. **Name**: Log Safety Incident. **Description**: "Logs a near-miss or incident against the shift and notifies the site safety officer."
3. Under **Completion**, respond using **IncidentReference**.
4. **Save**.

Original spec, for reference — Tool: Log Safety Incident. Type: Power Automate flow → Dataverse table 'SafetyIncidents' + Teams adaptive card. Logs a near-miss or incident against the shift, notifies the site safety officer, and creates an open action for follow-up. Inputs: Shift ID (from authenticated user's current shift); Incident type (Near-miss / First-aid / Reportable / Property damage); Location on site (string); Brief description (string, plain English); Immediate action taken (string). Outcome: Returns the incident reference number and posts an adaptive card to the site Safety Officer in Teams.

Configure, Evaluate, Test, Publish

Steps 7–10 — lock down settings, prove quality, and ship it.

7 Configure Settings

Settings cog (top right) — four areas before real users touch the agent:

- **General** — confirm name, description, primary language
- **Generative AI** — orchestration mode, content moderation (start at Medium)
- **Authentication** — "Authenticate with Microsoft" so the agent runs as the signed-in user
- **Security** — review DLP policy, enabled channels, end-user authentication

8 Evaluate Your Agent

Analytics → **Evaluations** → **New evaluation** — "Frontline Shift Handover Copilot — Baseline v1". Score every response on Groundedness, Relevance, Accuracy.

Input: Show near-misses at Site B in the last 14 days, grouped by equipment

Expect: Filters by Site=B, date within 14 days, sums NearMissCount, groups by equipment.

Input: What handover notes mention 'forklift'?

Expect: Returns rows where HandoverNotes contains 'forklift' (case insensitive) with shift IDs.

Input: Log a near-miss for my shift – slipped on a wet floor in the loading bay

Expect: Triggers the 'Log Safety Incident' tool with type Near-miss and location captured.

9 Test in the Test Pane

For each query, open **Activity map** to see: which topic/tool/knowledge fired, the inputs collected and outputs returned, citations to source rows, and per-step latency.

10 Publish to a Channel

Recommended for this build: Microsoft Teams (mobile) — frontline workers live in Teams on hand-held devices. Use **Teams mobile** as the primary channel for shift leads on the floor; **Microsoft 365 Copilot** for site managers reviewing trends from desktop; a **custom website** embed for the site intranet kiosk page; or **Telephony** (Azure Communication Services) for hands-busy voice handovers. Teams and M365 Copilot channels require tenant admin approval.

Try These Live

Sample queries to run once the agent is built — and what it replaces.

End of Shift

- Write my handover note for tonight's shift.
- Show open actions left by the previous shift lead.

Safety Trends

- Which site has the most near-misses in the last 14 days?
- Are any incident types recurring across sites this week?

Governed Action

- Log a near-miss — minor finger injury in the packing line.
- Try a paraphrase: 'someone tripped on a pallet, no injury'.

This agent replaces

- Paper handover sheets
- WhatsApp threads for near-miss reports
- Hand-built weekly safety roll-ups

With

- Structured handovers in under a minute
- Every incident captured with a reference number
- Site-level safety trends visible in real time

Tip for real-world use — publish to Teams mobile first — shift leads should never have to leave their hand-held device to log an incident. Add an approval branch so reportable incidents trigger immediate notification to the duty safety manager.